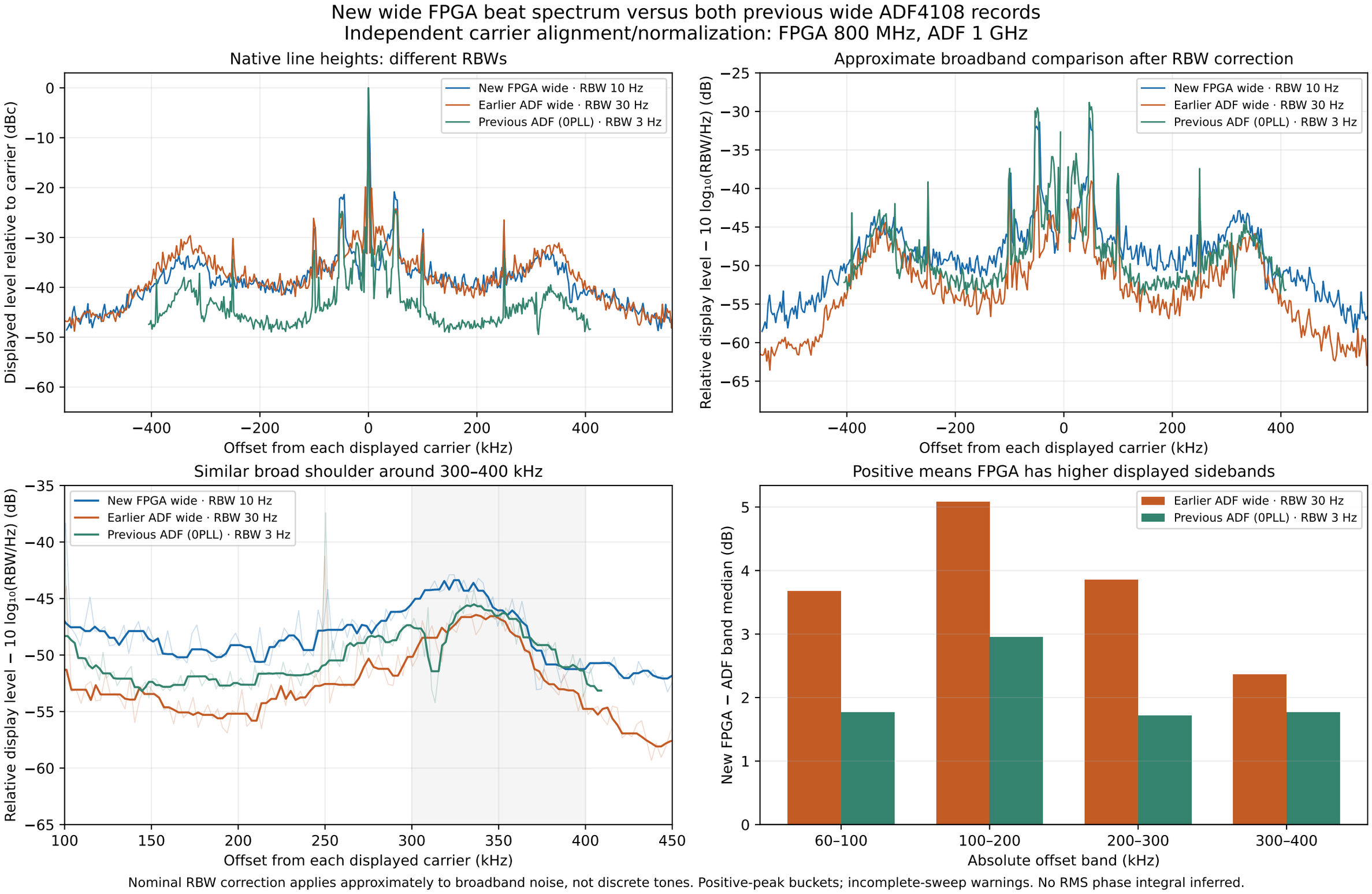

New wide FPGA beat spectrum versus both previous wide ADF4108 records
Independent carrier alignment/normalization: FPGA 800 MHz, ADF 1 GHz
Native line heights: different RBWs
Displayed level relative to carrier (dBc)
Offset from each displayed carrier (kHz)
New FPGA wide · RBW 10 Hz
Earlier ADF wide · RBW 30 Hz
Previous ADF (0PLL) · RBW 3 Hz
Approximate broadband comparison after RBW correction
Relative display level − 10 log₁₀(RBW/Hz) (dB)
Similar broad shoulder around 300–400 kHz
Positive means FPGA has higher displayed sidebands
New FPGA − ADF band median (dB)
60–100
100–200
200–300
300–400
Absolute offset band (kHz)
Nominal RBW correction applies approximately to broadband noise, not discrete tones. Positive-peak buckets; incomplete-sweep warnings. No RMS phase integral inferred.